MIL-P-80154A
28 April 1986
SUPERSEDING
MIL-P-80154
11 June 1968

MILITARY SPECIFICATION

PRESSES, PNEUMATIC, "C" FRAME, VERTICAL, ARBOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers pneumatic, "C" frame, vertical arbor presses of 3/4 ton through 18 tons capacity with equipment and accessories specified herein.

1.2 Classification. The presses shall be of the following types and of the sizes shown in tables I, II, and III. The type and size to be furnished shall be as specified (see 6.2.1).

Type I - Direct, single acting.

Type II - Direct, double acting.

Type III - Indirect, double acting.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government document. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Non-government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other information services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The press shall be new and one of the manufacturer's current models capable of pressing operations in accordance with the requirements herein. The press shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Dimensional systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the press. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurements, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.2 Reclaimed materials. The press may contain reclaimed materials provided such materials will not jeopardize the intended use and performance of the press. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the press.

3.2.3 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the press that present safety hazards. The safety devices shall not interfere with the operation of the press. The safety devices shall prevent unintentional contact with the guarded part and shall be removable to facilitate inspection, maintenance, and repair of the part. All press parts, components, mechanisms, and assemblies furnished on the press, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the press itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.5 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. All oil holes, grease fittings, and filler caps shall be accessible for maintenance.

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3.2.6 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or the press.

3.3 Construction. The press shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the press at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective component for use on the press. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the press shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the press and its related attachments and accessories shall conform to FED-STD-H28.

3.4 Components. The press shall consist of not less than a frame, air cylinder, ram, valves, necessary hoses, piping, tubing, fittings, and gauges.

3.4.1 Frame. The frame shall be of iron or steel and shall have a base with not less than four holes for either bench mounting or floor stand mounting. When specified (see 6.2.1), a floor stand shall be furnished for mounting the press. The bolster of the type I press shall be furnished with not less than two 1/2 inch bolt holes for mounting of the dies. The type II and III press bolsters shall be furnished with two 5/8 inch T-slots of the manufacturers standard length. The type I press frame shall be provided with vertical ways for raising and lowering the air cylinder and ram assembly.

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3.4.2 Air cylinder and ram assembly. The type I press shall be provided with a direct, single acting air cylinder and ram assembly. The air cylinder and ram assembly on the type I presses shall be vertically adjustable in conformance with the requirements of table I. Mechanical means shall be provided to raise and lower the air cylinder and ram assembly along the vertical ways. A means shall be provided to maintain the air cylinder and ram assembly in the adjusted position. The type II press shall be of the direct, double acting type. The type III press shall be of the indirect double acting type. Linkage from the air cylinder to the ram of the type III press shall be of sufficient strength and appropriate ratio to maintain performance characteristics in conformance with the requirements of table III. The ram of all type presses shall be provided with mechanical means, or mating threads on the ram nose, to securely restrain and seat a mechanical extension or adapter to the ram.

3.4.3 Valves. Unless otherwise specified (see 6.2.1), a directional control, hand operated, lever type valve shall be furnished to maintain control of the air cylinder and ram assembly. The directional control valve shall activate quick release type valves to prevent unnecessary delay in exhausting air from the air cylinder upon completion of the pressing cycle. Movements of the hand lever on the control valve shall infinitely control the amount of air pressure within the cylinder. Unless otherwise specified (see 6.2.1), inlet air pressure shall be immediately controlled by an air pressure regulator which is easily adjusted to the pressure and air volume consumption of the press.

3.4.4 Hoses, tubing, and piping. Hoses, tubing, piping, fittings, and their connections shall have a design safety factor of not less than 1.5 times their maximum operating pressure. Hoses shall not be adversely affected by oil or grease. All components shall be constructed in a manner to allow complete movement of any moving member of the press without modification of the air system.

3.4.5 Pressure gauges. The press shall be equipped with gauges to indicate both supply air pressure and regulated air pressure. Gauges shall be readable from the operator's normal operating position.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the press shall be not less than those specified in tables I, II, and III for the press size specified.

3.6 Performance. The press shall be capable of performing pressing operations within the rated capacity of the press, plus 10 percent overload, without permanent distortion or deformation to the press or the press components.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

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TABLE I. Type I, sizes and capacities.

Capacities	Sizes			
	1	2	3	4
Ram pressure, (lbs)	900	1800	2700	3800
Ram to table, ram up (inches)	8	8	12	12
Ram travel (inches)	2	2-1/2	3-1/2	4
Adjustment of ram height (inches)	5-1/2	5-1/2	8	8

TABLE II. Type II, sizes and capacities.

Capacities	Sizes			
	1	2	3	4
Ram pressure, (lbs)	900	2100	3800	6000
Ram to table, ram up (inches)	6	8	10	12
Ram travel (inches)	4	6	8	10
Reach, ram centerline to frame (inches)	6	8	10	12

TABLE III. Type III, sizes and capacities.

Capacities	Sizes									
	1	2	3	4	5	6	7	8	9	10
Ram pressure, (lbs)	1500	2500	4000	8000	10000	12000	16000	19000	25000	36000
Ram to table, ram up (inches)	9	9	19	19	19	19	19	19	23	23
Ram travel (inches)	4	4	5	6	4-1/2	6	4-1/2	6	6	6
Reach, ram centerline to frame (inches)	5	5	8	8	8	8	8	8	8	8

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3.7 Standard equipment. Unless otherwise specified, all equipment normally furnished with the manufacturer's standard commercial press shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.10 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.10.1 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each press. The nameplate shall contain the information listed below. If the press is a special model, the model designation shall include the model of the basic standard press and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Air input characteristics and ratings
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.12 Workmanship. Workmanship of the press and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturers designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The press shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The press shall be operated at no load for not less than 10 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Performance test. Each press shall be tested to determine compliance with its specified ram pressure rating. A ram extentions or adopter shall be installed and utilized in performing the test.

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4.7.3 Capacity test. Each press shall be tested with devices which measure and indicate the nominal pounds pressure rating specified for the size press to assure conformance to the requirements in section 3. Each press shall be subjected to an overload test which exceeds nominal pounds pressure rating by 10 percent. Following the overload test, the press shall be visually examined, without dismantling, for damage to components, adjustment, and alignment.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. These presses are primarily used for maintenance and repair work.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. First article, if required (see 3.1).

d. If press design and construction is required in a specific dimensional system (US or SI), state required system (see 3.2.1).

e. Additional safety and health requirements, if required (see 3.2.4).

f. Painting, if different (see 3.3.5).

g. Floor stand, if required (see 3.4.1).

h. Control lever, if different (see 3.4.3).

i. Air pressure regulator, if different (see 3.4.3).

j. Size and capacity, if different (see 3.5).

k. Additional performance features, if required (see 3.6.1).

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l. Standard equipment, if different (see 3.7).

m. Specify and fully describe optional equipment, if required (see 3.8).

n. Specify applicable repair parts, if required (see 3.9).

o. Nameplate, if different (see 3.10.1).

p. First article inspection, if different (see 4.4).

q. Quality conformance inspection, if different (see 4.5).

r. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.4 requires compliance only with those OSHA requirements that concern the press itself. It does not require compliance with those OSHA requirements that concern the press in its operating environment, such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. OSHA limits the total hazard level of these hazards in the environment and does not limit the hazard level of the individual press in the environment. The procuring activity is advised to analyze the existing hazard levels in the proposed operating environment and specify additional press requirements that will integrate the new press into its future operating environment. The press shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial press supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1 (e).

6.5 Subject term (key word) listing.

Arbor press
"C" frame press
Press, arbor, vertical
Press, pneumatic
Pressing operations

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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AR
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

User activity:
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Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-80174A	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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